

The Numbers

EasyCryptographypicoCTF 2019

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Description

The numbers... what do they mean?
[numbers.png](#)

Hints

1

The flag is in the format PICOCTF{}

116,641 users solved

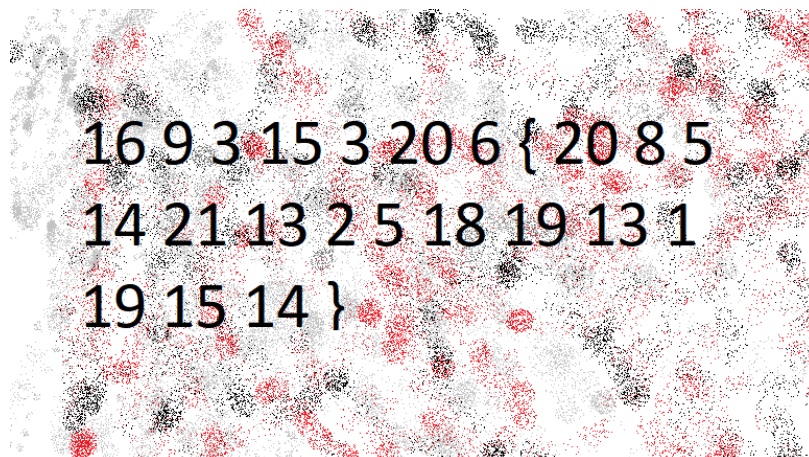
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This challenge, titled "The Numbers" from **picoCTF 2019**, is a classic entry-level cryptography puzzle. It focuses on a fundamental concept: alphabetic substitution based on numerical position.

Analysis

When you download and open the numbers.png file, you are presented with a sequence of numbers separated by spaces and braces:



Based on the hint provided in the challenge description, we know the flag format is PICOCTF{}. By comparing the number of digits before the bracket to the word "PICOCTF", we can see a direct correlation.

Decryption Strategy

The cipher used here is a simple A1Z26 cipher, where each number represents its corresponding letter in the alphabet:

- 1 = A
- 2 = B
- ...
- 26 = Z

Decoding the Prefix:

- 16 - P
- 9 - I
- 3 - C
- 15 - O
- 3 - C
- 20 - T
- 6 - F

This confirms the prefix is PICOCTF.

Decoding the Flag Content:

Now, we apply the same logic to the numbers inside the braces:

Number	Letter	Number	Letter
20	T	2	B
8	H	5	E

5	E	18	R
14	N	19	S
21	U	13	M
13	M	1	A
19	S	15	O
14	N		

The image shows a screenshot of the dCode website. On the left, a search bar contains the text 'e.g. type 'sudoku''. Below it, the results section shows 'Showing most likely results' with two entries: '[A1Z26] PICOCTFTHENUMBERSMASON' and '[A0B1Z25] QJDPDUGUIFOVNCFTSNBTPO'. A large blue button labeled 'Continue' is visible. On the right, the 'Letter Number Code (A1Z26)' tool is displayed. It includes a 'NUMBER TO LETTER A1Z26 CONVERTER' section with a text input field containing the sequence '16 9 3 15 3 20 6 20 8 5 14 21 13 2 5 18 19 13 1 19 15 14'. Below this is a 'DECRYPT AUTOMATICALLY' button. The 'MORE OPTIONS AND SETTINGS' section includes a 'CUSTOM ALPHABET' field with 'ABCDEFGHIJKLMNOPQRSTUVWXYZ', a 'CODE FOR 'SPACE' CHARACTER' field with '0', and a checkbox for 'THE MESSAGE HAS NO SEPARATOR BETWEEN NUMBERS'. A 'DECRYPT' button is at the bottom. A sidebar on the right contains a 'Summary' section with links to various tools and a 'Similar pages' section with links to 'Word's Value', 'Alphabetical Ranks Added', 'Consonants/Vowels Rank Cipher', 'Greek Letter Number α=1, β=2, γ=3', and 'T9 (Text Message)'.

Final Flag

Putting it all together, the sequence translates to:

PICOCTF{THENUMBERSMASON}

Note: The flag content is a reference to a famous line from *Call of Duty: Black Ops*—"The numbers, Mason! What do they mean?"